

Token-Efficient Context Sharing in Multi-Agent LLM Systems: How Much Can Be Cut, and What Limits It

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Abstract

Multi-agent LLM systems coordinate by forwarding context between agents, and every forwarded token is billed. Common practice forwards whole conversation histories, because deciding what to withhold is hard, so coordination cost grows quadratically in the number of agents. We ask a practical question: how much of that traffic can be removed without paying for it in task quality, and what stops us removing more?

On a controlled proxy for multi-agent context sharing built from a public multi-hop benchmark, a $2.6\times$ reduction in inter-agent tokens produces an uncertain 4.2-point accuracy loss (95% CI -10 to $+1$ points): the point estimate favours sending everything, and the interval does not establish a free region, only that both a real loss and no loss are compatible with this sample. An oracle reaches $22\times$ fewer tokens with an observed $+5$ -point *gain*, significant on the larger of two models tested. Over-delivery may thus be mildly harmful rather than merely wasteful, with roughly an order of magnitude available in principle, though the evidence for that ceiling is directional. A practical policy at $5.3\times$ establishes that a loss exists at that compression, 8 points with a CI excluding zero.

How the budget is spent matters more than how much is spent. Allocating per receiver beats selecting one globally-scored set for everyone by 19.2 accuracy points at matched cost; a separation result explains why: the one-set formulation, the natural seed-set analogy to influence maximization, loses a factor n in the number of receivers, by two independent constructions, even on monotone submodular instances, and adding saturation removes any finite bound. The two nearest optimisation toolchains do not apply here, the second because its structural parameters do not exist rather than because its bound is loose — yet the correct per-receiver formulation costs nothing extra in approximability, decomposing exactly into a resource-allocation program over a shared budget. Abandoning the one-set object is not a tractability trade.

Finally, a negative result that we think is the most useful finding for practitioners. None of three value-curve designs fed to a resource-allocation program that exploits structure no baseline can showed a detectable advantage over a tuned uniform per-receiver rule at matched budget, and replacing lexical relevance with a dense retriever makes matters worse. Three independent lines of evidence point to relevance *estimation* as the leading bottleneck on this proxy, not the allocation rule.

Keywords

multi-agent systems; large language models; communication; resource allocation; submodular optimization

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1 Introduction

A multi-agent LLM system coordinates by passing context. A planner tells a solver what the constraints are; an auditor is handed the records it must check. Every such transmission is billed per token, so coordination is a direct operating cost, and the prevailing engineering answer — forward the whole conversation, because deciding what to withhold is hard and sending everything is safe — makes that cost grow quadratically in the number of agents for architectures that accumulate or broadcast history (sparse topologies or prefix caching scale differently). We use transmitted tokens as a proxy for operating cost throughout; real billing also depends on input/output pricing, caching, and model choice.

This paper treats that as the problem to solve: how much inter-agent traffic can be removed before task quality suffers, what the cost–quality frontier’s shape is, and what prevents reaching cheaper allocations.

What we find. Three things, in decreasing order of how comfortable they are.

First, **cutting tokens produces an uncertain, not a zero, cost.** On a controlled proxy built from a public multi-hop benchmark, cutting inter-agent tokens by $2.6\times$ has an observed cost of 4.2 accuracy points (95% CI -10 to $+1$): the point estimate favours sending everything, and absent a pre-registered non-inferiority test we do not claim a free region (§6.1). An oracle allocation — which knows which item each agent needs — reaches $22\times$ fewer tokens with an observed $+5$ -point *gain*, significant on the larger of the two models we test, consistent with LLMs’ measured tendency to degrade under irrelevant context [18, 19, 25]. An order of magnitude is available in principle.

Second, **how the budget is spent matters more than how much.** Allocating a possibly different set to each receiver beats choosing one globally-scored set for everyone by 19.2 accuracy points at matched cost. This is not a tuning detail: the one-set formulation provably loses a factor n in the number of receivers (Proposition 1), by two independent constructions, on monotone submodular instances; once saturation is added, the same family admits no finite bound at all. That formulation is our seed-set analogy to influence maximization [15] (§3 states precisely where the analogy is and is not exact), and a substantial, otherwise applicable literature [1, 7, 23] optimises exactly that restricted object, which our separation shows is the wrong one here. We also show that



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the nearest general framework for the messy objective this problem actually has is unavailable, because its structural parameters fail to exist rather than merely giving a loose bound (§4.2). The switch of object costs nothing in return: the per-receiver problem decomposes exactly into a resource-allocation program coupled only by the shared budget, and any approximation available for one receiver transfers unchanged to n of them (Corollary 5).

Third, and most useful to anyone building such a system, **the remaining order of magnitude is not an optimisation problem.** We try three value-curve designs in the resource-allocation program, which exploits separability structure no per-receiver baseline can imitate, and none beats a tuned uniform per-receiver rule at matched budget. Replacing lexical relevance with a dense retriever makes per-receiver delivery *worse*. Three independent observations converge on relevance estimation, not the allocation rule, as this proxy’s leading bottleneck. In the composite-receiver design built to expose it (§5), the oracle reaches full delivery recall on 230 tokens where the best implementable policy reaches only 62% recall on 1050, and no allocation algorithm closes that gap, because the information required is absent from the scores rather than badly exploited.

Contributions. (i) A measured cost–quality frontier for inter-agent context sharing on a controlled proxy task, including an uncertain-cost token reduction and the oracle ceiling (§5). (ii) A separation result showing the one-set formulation loses a factor n even on monotone submodular instances and is unbounded once saturation is added (§4.1), with two degeneracy results removing the standard toolchains (§4.2), including a single instance failing both at once. (iii) An exact decomposition of the per-receiver problem into a resource-allocation program coupled only by the shared budget, with approximability equal to the single-receiver case under an integer-cost, oracle-access assumption stated precisely (§4.3). (iv) A negative result locating the residual gap in relevance estimation rather than allocation (§5). All theory statements were additionally checked by exhaustive enumeration on small instances.

2 Related work

Influence maximization. Monotonicity and submodularity of the spread function license greedy’s $(1 - 1/e)$ [15], and the scaling literature [1, 7, 23] inherits those assumptions. Recent work relaxes them but keeps the seed-set object: one chosen set, serving the whole network. Our separation is orthogonal to the quality of those relaxations.

Non-monotone and non-submodular maximization. Guarantees exist for non-monotone submodular maximization [2, 10], for weakly submodular objectives via the submodularity ratio [5, 6] or supermodular degree [9], and for differences of submodular functions [13]. Closest to our setting, Shi and Lai [22] treat non-monotone, non-submodular maximization under a knapsack; §4.2 shows their parameters do not exist here. Regularized objectives $g - c$ with g submodular and c modular admit distorted-greedy guarantees [12]; §4.3’s density rule is an instance of that lineage, credited as such rather than presented as new.

Context and communication cost in LLM agents. Empirical work cuts inter-agent tokens by pruning the message graph [26], generating the topology per task [8, 14], or sharing key–value caches instead of text [21], none with an optimality guarantee; a survey of token cost reports no formal treatment of allocation either [3]. Closest in spirit, [20] finds empirically that intermediate sparsity beats broadcasting, without an optimization formulation. Closest to our applied setting, RCR-Router selects a per-agent memory subset under a strict budget, up to 30% savings, again without guarantees [17]. A parallel line proves guarantees for a *single* receiver: PACMS treats context assembly as submodular selection [11]; BPS maximizes a monotone submodular benefit minus a modular token penalty, bicriteria $(1 - 1/e, 1)$, and is the closest prior work [4] — its benefit is a concave, dimension-additive coverage function, so a pair separately inert but jointly decisive has no representation in it, the case §4.2 turns on. [16] treats finite context windows as hard fan-in limits via a mixing depth, through majority-vote aggregation rather than set-function optimization. We are not aware of a formal treatment of multi-receiver token allocation, nor an influence-maximization framing of LLM-agent communication.

3 Model

Receivers (agents) are indexed $1, \dots, n$. A pool F of information items is given, item f costing $c(f) > 0$ tokens. An *allocation* assigns a set $S_i \subseteq F$ to each receiver. Information is copyable, but each transmission is charged, so total spend is $\sum_i c(S_i)$ and serving n receivers with one set costs n times serving one. We also consider a *free-broadcast* convention, where a set may be given to everyone for one transmission, to show our results do not rest on the cost model.

DEFINITION 1 (MULTI-RECEIVER CONTEXT ALLOCATION). *An instance is a tuple $(n, F, c, (u_i)_{i \leq n}, B)$ with n receivers, a finite item pool F , a cost $c : F \rightarrow \mathbb{R}_{>0}$ extended modularly to sets, per-receiver utilities $u_i : 2^F \rightarrow \mathbb{R}$ with $u_i(\emptyset) = 0$, and a budget $B > 0$. A feasible allocation is a tuple $(S_1, \dots, S_n) \in (2^F)^n$ with $\sum_i c(S_i) \leq B$, and the objective is $U(S_1, \dots, S_n) = \sum_i u_i(S_i)$. The seed-set restriction of the same instance requires $S_1 = \dots = S_n$.*

The seed-set restriction is our analogue, in a setting with no diffusion process, of the commitment an influence-maximization formulation makes: one chosen set as the entire decision variable. Classical IM’s diffusion can still *reach* different nodes differently once the seed set is fixed; here, where content is transmitted directly rather than propagated, the corresponding restriction is on the delivered content itself, which is why we state it as a restriction on Definition 1 rather than import an IM algorithm unmodified. Fixing everything else in Definition 1 and varying only this restriction is what makes the comparison in §4.1 a controlled one rather than a change of subject; §5’s seed-set baseline is our own translation of the idea (global top- k by aggregate score) rather than a published IM algorithm.

Per-receiver utility is

$$u_i(S) = \text{rel}_i(S) - \rho_i(\text{conf}_i(S)), \quad (1)$$

relevance minus a saturation penalty, with $\text{rel}_i(\emptyset) = \text{conf}_i(\emptyset) = \rho_i(\emptyset) = 0$ so that $u_i(\emptyset) = 0$ as Definition 1 requires. Two further features of (1) are empirically motivated rather than assumed

for convenience. First, ρ_i is increasing: delivering more context degrades an agent [19, 25]. Second, the penalty’s argument is *confusable* load conf_i , not raw token count, because length-matched studies find that semantic proximity to the task predicts degradation while raw length barely does [18]. Cost and damage are therefore different quantities in general: one is billed and modular, the other is semantic and per-receiver; §5 states explicitly where the experiments collapse this distinction for lack of a confusability oracle. The objective is $U = \sum_i u_i(S_i)$ subject to the budget.

Example 1. Three agents share a pool of 20 documents. Agent 1 must check a delivery date, agent 2 a payment total, agent 3 must reconcile the two. Agents 1 and 2 each need one document; agent 3 needs both. A uniform allowance of one document per agent starves agent 3; an allowance of two overfeeds agents 1 and 2 with a document that can only distract them. No single set serves all three, and the cheapest correct allocation sends different sets to different agents. This is the structure Proposition 1 makes precise.

Why routing intuitions do not transfer. The problem is priced like a delivery problem, which invites a vehicle-routing analogy that fails on one property: goods are conserved, so a delivered unit cannot serve a second customer, whereas context is *copyable* and one item can serve every receiver at once. Conservation is what makes routing a partition problem; without it, the interesting constraint is who should be *spared* an item rather than who gets the last one. Only *capacity* survives — a context window is rivalrous the way a vehicle is, which is what ρ_i encodes — and it is soft, per-receiver, and semantic, so a per-vehicle-capacity formulation is not a route to the multi-receiver guarantee we lack.

On the shape of ρ_i . We do not assume ρ_i convex: reported degradation curves are S-shaped, flat then a steep fall [18, 19], and magnitudes are strongly model-dependent (one study finds large models almost unaffected by non-confusable filler). Proposition 4 buys the freedom to avoid that fragility — the penalty enters only through a chosen budget level, so ρ_i may be non-convex, discontinuous, or a cliff. Only the density-greedy *heuristic* of §4.3, not the exact dominance test it approximates, needs ρ_i differentiable, and we flag that there as the heuristic’s limitation.

4 Theory: separation, degeneracy, and structure

4.1 The seed-set formulation loses a factor n

PROPOSITION 1. *For every $n \geq 1$ there is an instance with n receivers on which, at an identical budget,*

$$\frac{\text{OPT}_{\text{per-receiver}}}{\text{OPT}_{\text{seed-set}}} = n,$$

by two independent constructions: one from the hard budget alone (cost convention (a) below), one from saturation alone with the budget non-binding (convention (b)). For every $n \geq 2$, the same family of instances under convention (b) admits no finite multiplicative bound at all: the ratio grows without limit as a distraction penalty λ approaches a threshold from below, and at the threshold $\text{OPT}_{\text{seed-set}}$ is exactly 0 while $\text{OPT}_{\text{per-receiver}}$ stays at n .

PROOF. Take n receivers, n distinct items each of cost c , and let receiver i be satisfied only by item i . Set the budget $B = nc$.

(a) *Billed per receiver.* Per-receiver allocation sends item i to receiver i , spending $nc = B$ for utility n . A seed set S reaches all n receivers and so costs $n|S|c \leq nc$, forcing $|S| \leq 1$; a single item satisfies exactly one receiver, for utility 1. Ratio n .

(b) *Free broadcast, saturating attention.* Let the budget be non-binding, let a receiver absorb one item freely, and let each further item cost it 1. Broadcasting S to everyone yields $|S| - n \max(0, |S| - 1)$, maximized at $|S| = 1$ with value 1, while per-receiver allocation again yields n . Ratio n .

(c) *Unboundedness, $n \geq 2$.* Keep the same items and receivers, keep broadcast free, and let a receiver pay $\lambda \geq 0$ for each item it receives that is irrelevant to it. For a seed set S of size k , every item in S satisfies its own receiver and distracts the other $n - 1$, so summing over all n receivers gives $U_{\text{seed}}(S) = k(1 - \lambda(n - 1))$, independent of which k items are chosen. For $0 \leq \lambda < 1/(n - 1)$ this is increasing in k , so $\text{OPT}_{\text{seed-set}} = n(1 - \lambda(n - 1)) > 0$ and

$$\frac{\text{OPT}_{\text{per-receiver}}}{\text{OPT}_{\text{seed-set}}} = \frac{1}{1 - \lambda(n - 1)} \xrightarrow{\lambda \rightarrow (1/(n-1))^-} \infty,$$

so no λ -independent finite bound covers this family. At $\lambda = 1/(n - 1)$ the seed-set value is $U_{\text{seed}}(S) = 0$ for every S , so $\text{OPT}_{\text{seed-set}} = 0$; per-receiver allocation delivers item i to receiver i alone, incurs no penalty, and attains n regardless of λ . $\square$

Case (b) is the more informative one. A reader may discount (a) as an artifact of charging per receiver; (b) removes that objection, because even with free broadcast the seed set is capped. The reason is worth stating plainly: *an additional broadcast item helps one receiver and harms the other $n - 1$.*

Remark 1. The instance in case (a) contains no saturation and no complementarity: rel is modular and $\rho \equiv 0$, so the objective is monotone and submodular. The separation is therefore not a consequence of the degeneracies in §4.2. Repairing the objective’s curvature does not rescue the seed-set formulation, because the loss comes from the choice of object, not from the objective’s shape. Any guarantee proved against a seed-set optimum bounds distance from the wrong quantity.

Neither construction claims to be the worst case over any class, and we make no such claim: an instance where every receiver needs the same item gives ratio 1 under either convention, so the seed-set loss is not uniformly n across monotone submodular instances, only *at least* n on some of them. What (a)–(c) together show is an existence result on two fronts: the gap is already n on instances as well-behaved as one could ask (modular relevance, no saturation, so U is monotone and submodular by Proposition 3’s own terms), and once saturation is strong enough to make broadcasting actively harmful, the same family admits no finite multiplicative bound at all. Both constructions were checked by exhaustive enumeration over small instances rather than by algebra alone.

For completeness we record the hardness of the correct object; it is routine and we claim no credit for it.

PROPOSITION 2. *Maximising U subject to $\sum_i c(S_i) \leq B$ is NP-hard, already for $n = 1$ and $\rho \equiv 0$.*

PROOF. With one receiver and no saturation the problem is to choose $S \subseteq F$ with $c(S) \leq B$ maximising a modular rel , which is KNAPSACK. $\square$

Remark 2. General n does *not* add MULTIPLE KNAPSACK’s structure (per-bin capacities, no reuse across bins): here every receiver draws against *one* shared B and items are copyable, so treating each receiver-item pair (i, f) as its own object reduces general n (with rel modular) to ordinary KNAPSACK over a larger ground set — no harder than $n = 1$, consistent with §3’s point that conservation is what does not transfer.

Remark 3. What Proposition 2 does *not* settle is whether saturation adds hardness beyond the knapsack structure. The construction above switches ρ off, so the hardness is entirely attributable to the budget. We leave the question open and note that Proposition 5 below is evidence in the other direction: the multi-receiver structure adds no approximation hardness at all.

4.2 Both toolchains degenerate

Definition 1 makes an allocation a tuple $(S_1, \dots, S_n) \in (2^F)^n$, one set per receiver, but submodularity and the two parameters below are properties of a *set* function, made explicit once here rather than left implicit. Let $E = [n] \times F$ be the receiver-item ground set; identify an allocation with $A = \{(i, f) \in E : f \in S_i\} \subseteq E$, and define $U(A) = \sum_i u_i(\{f : (i, f) \in A\})$, agreeing with Definition 1’s objective. 2^E is the domain on which (2)–(3) are stated below, and the witnesses in Proposition 3 are sets of receiver-item pairs in it; the combined instance’s two violating triples occupy disjoint coordinates of E (different receivers), which is why both survive side by side.

PROPOSITION 3. *Neither hypothesis of the greedy $(1 - 1/e)$ argument, nor either parameter of [22]’s general theorem, holds across the class of Definition 1. Precisely, on two separate witnesses:*

(a) Saturation. *There is an instance with rel modular and ρ increasing on which U is not monotone, and no $\gamma_2 \geq 1$ makes U γ_2 -weak supermodular in the sense of [22].*

(b) Complementarity. *There is an instance — necessarily with rel not modular on the receiver where it is evaluated, since a modular rel cannot satisfy the complementarity below — on which U is not submodular, and no finite γ_1 makes U γ_1 -weak submodular.*

The two witnesses can in fact be forced simultaneously, on a single instance: give one receiver (a)’s items and a second, disjoint receiver (b)’s items. Since neither witness’s violating triple involves the other receiver’s items, both violations survive intact in the combined instance, so no finite γ_1 and no $\gamma_2 \geq 1$ exist for it at once. [22]’s Theorem 4 therefore does not merely fail to cover some instance of Definition 1 — it fails on instances that need only two receivers.

Monotonicity and γ_2 fail under saturation. With rel modular and ρ increasing, an item whose relevance falls below its marginal saturation cost strictly lowers u_i , so U is not monotone; this is witness (a).

Submodularity and γ_1 fail under complementarity. Two items individually inert but jointly decisive, with $\rho \equiv 0$: $\text{rel}(\emptyset) = \text{rel}(\{a\}) = \text{rel}(\{b\}) = 0$, $\text{rel}(\{a, b\}) = 1$ — give increasing returns, so U is not submodular; this is witness (b). Definition 1 places no modularity requirement on rel_i , so this instance is a legitimate member of the class even though it falls outside witness (a)’s hypothesis. Both failures are required by the greedy argument, on their respective instances.

Table 1: Existence of the two parameters that [22] requires simultaneously. Each phenomenon destroys exactly one.

	γ_1 (weak sub.)	γ_2 (weak super.)
Complementarity	does not exist	exists
Saturation	exists	does not exist

Witness (a): one receiver, $F = \{a, b\}$, modular $\text{rel} = \text{conf} = |S|$, convex saturation $\rho(0)=0$, $\rho(1)=0.5$, $\rho(2)=3$. Then $u(\{a\}) = 0.5$ and $u(\{a, b\}) = -1$, so the marginals (3) is checked against are $+0.5$ then -1.5 . Both witnesses and the combined instance are verified exhaustively in `code/degeneracy_check.py`.

The nearest general framework has no parameters here. Shi and Lai [22] give guarantees for non-monotone, non-submodular maximization under a knapsack. Their general theorem requires the objective to be simultaneously γ_1 -weak submodular and γ_2 -weak supermodular:

$$F(U_2 \cup \{x\}) - F(U_2) \leq \gamma_1 [F(U_1 \cup \{x\}) - F(U_1)], \quad (2)$$

$$\gamma_2 (F(U_2 \cup \{x\}) - F(U_2)) \geq F(U_1 \cup \{x\}) - F(U_1), \quad (3)$$

for $U_1 \subseteq U_2$ and $x \notin U_2$. Our two phenomena destroy exactly one parameter each.

Under complementarity, take the instance above with $U_1 = \emptyset$, $U_2 = \{b\}$, $x = a$: (2) demands $1 \leq \gamma_1 \cdot 0$, which no finite γ_1 satisfies. Under saturation, an item whose marginal is $+0.5$ at a lightly loaded receiver and -1.5 at a saturated one makes (3) demand $\gamma_2 \cdot (-1.5) \geq 0.5$, which no $\gamma_2 \geq 1$ satisfies.

Table 1 summarizes, and the witnesses above — including the combined two-receiver instance, checked by exhaustive search over its whole ground set rather than by algebra alone — are its proof. This is a stronger statement than a loose bound: the required parameters need not exist for instances in the class, not merely take large values, and a single small instance can lack both at once. It does not say [22] is wrong: its parameters are exactly right for the class it addresses, most instances of Definition 1 (any modular rel with $\rho \equiv 0$, for one) admit both, and a framework covering ours would need unbounded local curvature in both directions, which we do not know how to build. Nor is the result deep: Definition 1 places essentially no structure on rel_i , so *some* instance escaping any fixed finite parameterisation is unsurprising. What we take from Proposition 3 is narrower and, we think, still useful: saturation and complementarity are two concrete, independently occurring failure modes, not a claim that this class is uniquely pathological among unrestricted set functions.

4.3 Structure and algorithms

Proposition 1 says the per-receiver object is the right one. It does not say the resulting problem is intractable, and in one respect it is unusually well structured.

PROPOSITION 4. *The problem is separable across receivers and coupled only by the shared budget. Consequently, writing $v_i(b) = \max\{u_i(S) : c(S) \leq b\}$, an optimal allocation is obtained by solving*

$$\max \left\{ \sum_i v_i(b_i) : \sum_i b_i \leq B \right\}$$

and giving receiver i an optimal set at its level b_i . If each v_i is computed exactly, this is exact.

PROOF. $U = \sum_i u_i(S_i)$ and the only constraint linking receivers is $\sum_i c(S_i) \leq B$. Fix any feasible allocation and let $b_i = c(S_i)$. Then $u_i(S_i) \leq v_i(b_i)$ and $\sum_i b_i \leq B$, so its value is at most that of the displayed program. Conversely any feasible (b_i) with maximisers S_i is a feasible allocation of value $\sum_i v_i(b_i)$. $\square$

Two consequences matter. First, the coupling between receivers is a single scalar variable, the shared budget B , and nothing else — no pairwise interaction between receivers survives the reformulation. Second, and less obviously, the saturation penalty enters *only* through the chosen level b_i , so ρ_i may have any shape whatever — non-convex, discontinuous, a cliff. This matters because the measured degradation curves are S-shaped rather than convex, and a rule that reasons about ρ'_i requires convexity to argue that its admission bar rises as a receiver fills.

Remark 4. The scalar coupling is *not*, in general, a market-clearing price: that needs every v_i concave, which Proposition 4 does not assume. Take $v_1(0, 1, 2) = (0, 0, 3)$, $v_2(0, 1, 2) = (0, 2, 2)$, $B = 2$: the optimum spends all of B on receiver 1, value 3. For $\lambda < 1.5$ the local optima $\arg \max_b v_i(b) - \lambda b$ sum past B (infeasible); for $\lambda \geq 1.5$ they sum to value at most $2 < 3$. No scalar λ implements the optimum; Algorithm 1 is exact here precisely because it does not route through a price. What transfers to every v_i , concave or not, is the reformulation and Corollary 5's guarantee — not a market interpretation of the multiplier.

The decomposition also yields the one positive guarantee we have. It says that the difficulty of the multi-receiver problem is concentrated entirely in the single-receiver subproblems: whatever can be achieved for one receiver transfers, without loss, to n of them.

COROLLARY 5. *This is a computational oracle in the algorithmic sense, unrelated to the gold-support policy Table 3 calls “oracle”; suppose for each receiver i we have one that, given a level b , returns a set $\tilde{S}_i(b)$ with $c(\tilde{S}_i(b)) \leq b$ and $u_i(\tilde{S}_i(b)) \geq (1 - \varepsilon) v_i(b)$. Then Algorithm 1, run on the curves $\tilde{v}_i(b) = u_i(\tilde{S}_i(b))$, returns a feasible allocation of value at least $(1 - \varepsilon) \text{OPT}$.*

PROOF. Let (b_i^*) maximise $\sum_i v_i(b_i)$ subject to $\sum_i b_i \leq B$; by Proposition 4 its value is OPT. The algorithm returns $\max\{\sum_i \tilde{v}_i(b_i) : \sum_i b_i \leq B\}$, which is at least $\sum_i \tilde{v}_i(b_i^*) \geq (1 - \varepsilon) \sum_i v_i(b_i^*) = (1 - \varepsilon) \text{OPT}$. Feasibility and attainment hold because each $\tilde{v}_i(b)$ is realised by an actual set of cost at most b , and $\tilde{v}_i(b) \geq 0$ since $\emptyset$ is always available. $\square$

Remark 5. The technique is standard resource allocation, credited as such, under two standing assumptions. Costs are integer, so running time (Algorithm 1's table is indexed by $b = 0, \dots, B$) is pseudo-polynomial in B , not polynomial in its bit length; and the oracle hypothesis must hold at *every* level b . With $\rho_i \equiv 0$ and rel_i additionally modular, each v_i is an ordinary knapsack value function with a $(1 - \varepsilon)$ -oracle for every ε ; for general monotone submodular rel_i , the standard value-oracle guarantee is $(1 - 1/e)$, and the Corollary transfers that coarser ratio instead. Either way the n -receiver problem is no harder to approximate than the 1-receiver one: the

Algorithm 1 Marginal allocation, given any per-receiver oracle

Require: receivers $1..n$, pool F , integer costs, budget B , and an oracle returning $(V_i(b), S_i(b))$ for every i, b — exact ($V_i = v_i$, Proposition 4) or $(1 - \varepsilon)$ -approximate ($V_i = \tilde{v}_i$, Corollary 5)

```

1: for  $i = 1$  to  $n$  do
2:   for  $b = 0$  to  $B$  do
3:      $(V_i(b), S_i(b)) \leftarrow \text{oracle}(i, b)$ 
4:   end for
5: end for
6:  $A_0(b) \leftarrow 0$  for all  $b$ 
7: for  $i = 1$  to  $n$  do
8:    $A_i(b) \leftarrow \max_{0 \leq t \leq b} A_{i-1}(b - t) + V_i(t)$   $\triangleright$  for all  $b \leq B$ 
9: end for
10: recover the levels  $(b_i)$  attaining  $A_n(B)$  return
     $(S_1(b_1), \dots, S_n(b_n))$ 

```

seed-set formulation is off by a factor n or worse (Proposition 1) while the correct one costs nothing extra, so abandoning it is not a tractability trade.

Example 2. In the running example, v_1, v_2 jump $0 \rightarrow 1$ at $b = 1$; v_3 stays 0 until $b = 2$, then jumps to 1. At $B = 4$ the DP splits $(1, 1, 2)$ for value 3; uniform $(b_i = 4/3, \text{rounding to } 1 \text{ each})$ scores only 2, wasting a document agent 3 cannot use alone. Greedy density also scores 2: agent 3's first document has marginal value 0 and loses to agent 1's or agent 2's marginal 1, and a zero-marginal item once passed over is never revisited — the DP avoids this because v_3 is built in full before any budget commits.

Concavifying v_i (time-sharing between two integer levels) does restore a market-clearing price, but only in *expectation*: if each receiver randomizes independently, the realized total spend can exceed B in a given draw even though $\mathbb{E}[\sum_i b_i] = B$. Exact ex-post feasibility needs the randomization correlated across receivers (e.g. dependent rounding), not independent per-receiver coin flips; we do not work out that construction here.

The second loop is $O(nB^2)$ arithmetic operations, or $O(nB)$ with the usual running-maximum bookkeeping when the v_i are concave; either way it is pseudo-polynomial in the budget and linear in the number of receivers, so the coupling across receivers is not where the cost lies. The first loop is the hard part, by Proposition 2. This is the sense in which the per-receiver object is well structured: the multi-receiver coupling is a scalar and the difficulty is local.

Two algorithm families follow, and we are explicit about what is new.

An exact dominance test. No optimal allocation contains an item $f \in S_i$ with

$$\Delta \text{rel}_i(f | S_i \setminus \{f\}) < \rho_i(\text{conf}_i(S_i)) - \rho_i(\text{conf}_i(S_i \setminus \{f\})). \quad (4)$$

This is exact, not a first-order approximation: if (4) holds, deleting f raises u_i , because the relevance lost is smaller than the saturation charge *exactly* avoided, and simultaneously frees $c(f) > 0$ of budget, which can only weaken the constraint. The resulting allocation is feasible and strictly better, contradicting optimality, and this needs no assumption on ρ_i 's shape — not even continuity. It is, however, a condition on the *completed* S_i , checked *ex post*: both sides depend on S_i itself, so it is a necessary property of an optimum,

not a rule for admitting items one at a time as a set is built. Under complementarity in particular, an item that fails (4) against a smaller intermediate set can satisfy it once the rest of S_i is in place, so a forward construction that rejects on first failure can discard an item an optimum would keep.

A density-greedy heuristic. Building a set that satisfies (4) still requires deciding an admission *order*. The standard device, which we credit rather than claim (distorted-greedy [12], Algorithm 1 of [22], applied to a single agent with a bicriteria guarantee by [4]), ranks candidates by relevance per token, $\Delta \text{rel}_i(f \mid S)/c(f)$, and admits while a threshold is cleared. Turning (4) into a per-token threshold requires relating the exact penalty step to $c(f)$, and this is where the model’s own distinction between billed cost and confusable load (§3) has to be specialized to say anything operational: writing $\text{conf}_i(S) := c(S) - \text{confusable load approximated by token load}$, for lack of a confusability oracle — turns (4) into a same-unit condition on $c(S_i)$, and a first-order approximation of *that* gives the per-token rule $\Delta \text{rel}_i(f \mid S)/c(f) \geq \rho'_i(c(S))$ our density-rule baseline (§5) implements. Two approximations are stacked here, not one: the token-load proxy for confusability, and the first-order step, which additionally needs ρ_i differentiable where the measured curves are S-shaped. We report the baseline’s results as a heuristic under both approximations, not as an instantiation of the exact test. A per-receiver threshold, exact or approximate, also has no view of what a token would buy at another receiver — which is exactly what Algorithm 1 adds.

Marginal allocation. The objective is separable across receivers and coupled only by the shared budget, so the problem factors: build each receiver’s value-versus-budget curve $v_i(b)$ and solve $\max \sum_i v_i(b_i)$ subject to $\sum_i b_i \leq B$ by dynamic programming. This equalizes marginal value per token *across* receivers, which neither a uniform per-receiver rule nor a per-receiver density rule can do — the former spends identically on every receiver, the latter has no view of what a token would buy elsewhere. The saturation penalty enters only through the chosen level, so it may take any shape, including the non-convex ones the measurements suggest. §5 reports that this does not, in fact, outperform a tuned uniform rule on our benchmark, and why.

5 Experiments

Setup. We use MuSiQue-Ans [24], where the structure we need is annotated: each instance supplies 20 candidate paragraphs with support labels, and sub-questions each carrying the index of the paragraph answering it. Receivers are sub-questions, the item pool is the paragraph set, and cost is tokens transmitted summed over receivers. Ground-truth relevance being labeled makes an oracle allocation computable. As §4.3 flags, we take $\text{conf}_i(S) := c(S)$ throughout, i.e. token load stands in for confusable load; utility is proxied by exact-match accuracy, since rel_i and ρ_i are not separately observable outcomes. Neither substitution is assumed elsewhere: every theory result in §4.1–§4.3 holds for general conf_i and u_i .

Splitting MuSiQue’s single-agent task across per-sub-question receivers is our framing, discussed as a threat to validity in §6.1; we restrict to branching families where two sub-questions are genuinely independent. Comparisons are paired — every policy answers

the same items — using McNemar’s exact test on discordant pairs and a bootstrap over items.

Protocol. `gemini-3.5-flash-lite` (matched-budget grid) and `gemini-3.1-flash-lite` (the earlier grid), both at temperature 0, called September 2026, sharing one prompt template (shuffled paragraphs, then the question, “reply UNKNOWN” if absent, answer on a line marked ANSWER:) and one extraction rule (text after ANSWER:, else the reply’s last line), so formatting cannot differ between policies and arms differ only in which paragraphs the receiver is given. Lexical relevance is normalized bag-of-words overlap, $|q \cap t|/\sqrt{|q||t|}$ over non-stopword tokens; the embedding scorer is `gemini-embedding-001` at 768 dimensions, cosine similarity. Token counts reported everywhere here are real countTokens responses on the exact transmitted context, not a word-count proxy; selection is deterministic given the instance and policy, so the exact context for every logged call is reconstructed and recounted rather than re-estimated (`code/real_tokens.py`), excluding the fixed per-call scaffold (54 tokens, identical across every arm). Temperature 0 was verified to reproduce identical output across repeated calls, and every response’s `modelVersion` field confirms the named model. $q \in \{0.3, 0.7, 0.95\}$ are quantiles of a receiver’s own candidate-density distribution (§4.3’s `auto_scale`); $k \in \{3, 5\}$ are the cardinalities reported; neither was chosen on a calibration split disjoint from the evaluation grid — we do not have one, so we cannot certify “tuned uniform” is free of having been examined on overlapping data (§6.1). The matched-budget grid has 12 arms over 60 instances (1440 calls, all completed) on `gemini-3.5-flash-lite`; on `gemini-3.1-flash-lite` a 12-arm, 19-instance pilot and a separate 4-arm, 125-instance confirmatory grid were run, both the full branching-family split, no further filtering. One instance’s run in the confirmatory grid was cut short by API quota and is excluded from every paired comparison that uses it, leaving 124 instances (248 receiver-instances) with complete data on all four arms. Arms are matched on delivered tokens before accuracy is compared, since an accuracy gain bought with a larger budget is not a gain from the mechanism. Runs are keyed and resumable per (instance, policy, arm, receiver), and duplicate keys are removed before analysis; without this the pairing silently breaks and n inflates.

The separation appears in the data. The strongest measured effect is the first row of Table 2: per-receiver allocation beats a seed-set baseline — items ranked by aggregate score and sent to everyone — by 19.2 accuracy points at matched cost. Proposition 1 motivates the hypothesis that this gap grows with the number of receivers — it is an existence result, not a prediction about this fused construction, where the candidate pool and ranking competition also grow with n — and the hypothesis holds here: as we fuse instances to raise n from 2 to 8 to 16, seed-set delivery recall falls from 0.45 to 0.08 to 0.01 while per-receiver recall stays above 0.55.

A cost–quality frontier, honestly reported. The loose density rule cuts tokens 2.6× at an observed, uncertain −4.2-point loss, 95% CI [−10, +1] points (row 3): the point estimate favours broadcast, and most of the interval is negative, so this does not establish a free region — only that a loss up to 10 points and a loss of zero are both compatible with the data. We report the estimate and interval

Table 2: Paired comparisons, $n = 120$ receiver-instances (60 instances $\times$ 2 independent sub-questions), matched-budget grid. Δ is the first policy minus the second; tokens are each arm’s mean, real countTokens counts on the exact transmitted context, not a word-count proxy (broadcast 2517; code/real_tokens.py). CI and p are the *primary*, cluster-robust statistics from code/cluster_test.py: a block bootstrap and a permutation test resampling the 60 source instances, not the 120 receivers, since each instance contributes two non-independent receivers — the flat, receiver-level McNemar test differs negligibly (largest shift: this p 0.013 $\rightarrow$ 0.029) and is not shown. Starred $p < 0.05$.

Comparison	Δ acc	95% CI	p	tokens
per-r. $k=3$ (254) vs. seed $k=3$ (283)	+0.192	[+.10, +.28]	0.0003*	1.1 $\times$
oracle (115) vs. b’cast	+0.050	[−.01, +.11]	0.140	21.9 $\times$
density, loose (952) vs. b’cast	−0.042	[−.10, +.01]	0.225	2.6 $\times$
unif. $k=5$ (478) vs. b’cast	−0.083	[−.15, −.02]	0.027*	5.3 $\times$
density, tight (586) vs. unif. $k=5$	+0.000	[−.05, +.05]	1.000	0.8 $\times$

Table 3: Accuracy and cost per policy, matched-budget grid, $n = 120$. Tokens are per receiver, real countTokens counts (not a word-count proxy); the ratio is broadcast’s mean tokens divided by the policy’s, from the two numbers shown. Absolute exact-match is suppressed for deltas since MuSiQue predates current models (§6.1: 0% no-context accuracy here, up to 5% on the other model’s pilot). “Density” is §4.3’s density-greedy heuristic (an approximation of the exact dominance test, *not* an instance of it) at two thresholds; it is a different policy from Algorithm 1, evaluated separately in Table 4.

Policy	tokens	ratio	Δ acc
broadcast (all items)	2517	1.0 $\times$	—
oracle (labelled)	115	21.9 $\times$	+0.050
per-recv., unif. $k=3$	254	9.9 $\times$	−0.200
per-recv., unif. $k=5$	478	5.3 $\times$	−0.083
density, tight $q=0.7$	586	4.3 $\times$	−0.083
density, loose $q=0.3$	952	2.6 $\times$	−0.042
seed set, $k=3$	283	8.9 $\times$	−0.392
random, $k=3$	390	6.5 $\times$	−0.667
no context	0	—	−0.792

rather than a verdict; we did not run a pre-registered non-inferiority test. Pushing to 5.3 $\times$ establishes that a loss exists at that stronger compression, 8.3 points with a CI excluding zero (row 4); it does not by itself trace the frontier’s shape between the two points. The oracle beats broadcast by an observed ~ 5 –7 points at 22 $\times$ fewer tokens, consistent in sign across two models and significant on the larger of the two (+0.065, $p = 0.001$, $n = 124$) though not on the primary grid (+0.050, $p = 0.140$, $n = 60$) — a real effect, not yet established at the primary grid’s own sample size.

A negative result on algorithm design. Across three value-curve designs and both conditions (Table 4), no curve beats tuned uniform at matched nominal spend: sum-of-scores loses on both (CI excludes zero); noisy-OR ties on both (CI crosses zero); captured-mass loses on plain but ties once composite receivers add heterogeneity — the one cell not detectably worse than uniform. Nor does the density rule (Table 2, row 5, exactly zero at $p = 1.000$ while spending 23% more). We also found that replacing lexical relevance with one

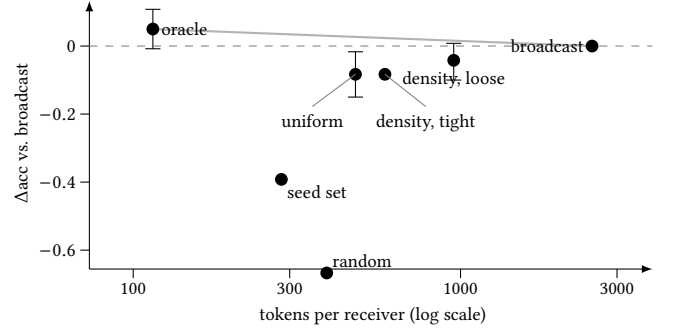


Figure 1: The measured cost–quality frontier, $n = 120$ receiver-instances, real countTokens counts. Vertical bars are 95% cluster-robust CIs for the three points Table 2 reports against broadcast (oracle, density-loose, uniform $k=5$); the other points have no CI reported here. The dashed line is broadcast’s accuracy; anything above it is both cheaper *and* better. None of the three plotted intervals clears zero on the positive side: oracle’s and density-loose’s both cross the dashed line, and uniform $k=5$ ’s interval excludes zero entirely but on the *negative* side, i.e. is confidently below broadcast, not above it. The implementable policies cluster well below the grey envelope, and — the point of the paper’s negative result — they differ from one another by far less than any of them differs from the oracle. The seed set at matched cost is 39 points down, and random delivery (−0.667) fixes the scale of the task. An empty-allocation arm (not shown, 0 tokens, −0.792) is the contamination control.

Table 4: Three designs for $v_i(b)$: $\sum \text{raw (sum)}$, $1 - \prod (1 - \text{raw})$ (noisy-OR), $\sum \text{raw} / \sum \text{raw}$ (captured mass, Algorithm 1’s own curve). Scored by delivery recall (deterministic) at matched nominal spend against tuned uniform, on $n = 130$ branching instances (plain, 260 receivers, uniform recall 0.673) and the same 130 with a composite receiver added (390 receivers, uniform recall 0.528). Δ is curve minus uniform with a 95% cluster bootstrap CI over instances; tokens are real countTokens counts (code/value_curves_check.py), freshly computed since the code behind a prior version of this table no longer exists (see STATUS.md).

Curve	Δ [95% CI], plain	Δ [95% CI], composite	tokens (pl./comp.)
sum	−0.092 [−.14, −.05]	−0.044 [−.07, −.02]	225/420
noisy-OR	−0.012 [−.05, +.02]	−0.013 [−.03, +.01]	220/415
captured	−0.046 [−.08, −.01]	+0.003 [−.02, +.02]	221/416

(unnamed) dense embedding scorer makes per-receiver retrieval worse (0.43 $\rightarrow$ 0.25 top-1 recall) while nearly doubling seed-set recall — a suggestive contrast we discuss below, not yet a causal one.

Composite receivers — a third receiver added to each of 130 branching instances from the same pool as the two grids above, holding the instance’s full multi-hop question, which needs *all* its supporting paragraphs where a sub-question receiver needs one — manufacture the heterogeneity Proposition 1 draws its advantage from, so a global k must starve one receiver or overfeed the other. Even then, marginal allocation only matches uniform: it can only

Table 5: Replacing lexical relevance with a dense embedding scorer *harms* per-receiver retrieval but helps the seed set. The two signals fail in opposite directions: embeddings capture what an instance is about, lexical matching captures which receiver needs what.

Delivery recall	lexical	embedding
per-receiver, top-1	0.43	0.25
per-receiver, top-5	0.79	0.61
seed set, $k=3$	0.45	0.85

help where per-receiver value curves are distinguishable from label-free signals, and here they are not, since every receiver draws from a common pool with a similar score distribution. Heterogeneity in what receivers *need* did not surface as heterogeneity in what could be *estimated*.

A plausible, unconfirmed direction: MuSiQue’s branching sub-questions are mostly terse relation templates, not natural-language questions, so exact entity-token overlap locates the right paragraph directly. The seed-set row of Table 5 moves the other way ($0.45 \rightarrow 0.85$), suggestively not causally, since one global set needs only instance-level topicality. We hypothesize the residual gap partly reflects query representation, not scorer family; a hybrid or entity-aware scorer, tested against a named set of retrievers, is the obvious next attempt. We have not tried it.

Figure 1 puts all of this on one axis. The visual summary is the paper’s argument in miniature: the gap between the implementable policies and the oracle dwarfs the gaps among the policies themselves, consistent with relevance estimation rather than the allocation rule being what limits these results.

Three independent observations thus point the same way: on this proxy, relevance *estimation* looks like the leading bottleneck, not allocation *optimization*. On the composite-receiver design just described, the oracle fixes the scale of it, reaching full delivery recall on 230 tokens where the best implementable policy reaches only 62% recall on 1050 — a different sample and metric from Table 3’s accuracy grid, and not directly comparable to its 115-versus-broadcast tokens. No allocation algorithm closes this gap, because the information required is not present in the scores.

Logged traces. Two individual calls from the matched-budget grid, tracing an over-delivery failure and a seed-set failure item by item down to the model’s own answer, are in the supplementary material rather than here, to leave room for the analyses above.

6 Discussion and conclusion

6.1 Discussion

The results divide cleanly. The formulation-level claims are strong: the seed-set object is wrong by a factor n on monotone submodular instances, unboundedly wrong once saturation is added, and both the classical toolchain and its nearest general successor are unavailable here. The algorithm-level claims are deliberately weak: no tested method beats tuned uniform, reported rather than a favorable budget point — informative in itself (§5: heterogeneity in what receivers *need* did not surface as heterogeneity in what could

be *estimated*). Genuinely different, predictable appetites are where the separation’s practical value should be sought.

Threats to validity. We list these in the order we think a referee should weigh them.

The multi-receiver framing is ours. Splitting MuSiQue into per-sub-question receivers is defensible — sub-questions and supporting paragraphs are annotated, making the oracle computable — but no conclusion here measures a deployed multi-agent system; we restrict to branching decompositions since sequential ones would make receivers dependent.

Contamination. The benchmark predates current models: 0% of receiver-instances in Table 3’s run answer correctly with no context, rising to 5% on the other model’s smaller 19-instance pilot (not rerun at the confirmatory grid’s scale). We report policy deltas, not absolute exact-match, so memorisation cannot be mistaken for policy quality if it affects arms equally.

Statistical power, clustering, and cross-model scope. Only oracle-vs-broadcast replicates on the second model (gemini-3.1-flash-lite, a different sample) — $+0.065$ there (cluster-robust $p = 0.001$, $n=124$) vs. $+0.050$ on the primary grid ($p = 0.140$), consistent in sign and significant on one of the two; the headline per-receiver-vs-seed-set result and Table 4 run on gemini-3.5-flash-lite alone. Each instance also contributes two or three non-independent receivers, so Table 2 resamples and permutes at the instance level; the unreported receiver-level version differs negligibly (three of five rows unchanged to three decimals, largest shift $p: 0.013 \rightarrow 0.028$).

Calibration of p . The saturation penalty’s shape comes from published degradation curves whose magnitude is model- and task-dependent, and at least one study finds large models unaffected by non-confusable filler. Proposition 1(a) needs no degradation at all; the over-delivery claim is only as general as that literature.

Open problems. (i) An unconditional guarantee for non-trivial p_i (Corollary 5 reduces this to the non-monotone, non-submodular single-receiver case), and whether saturation adds hardness beyond Proposition 2’s knapsack structure. (ii) Bundle identification: our witness assumes the jointly-decisive pair is known, not discovered. (iii) An end-to-end multi-agent task on a dependent graph, against stronger baselines. (iv) Endogeneity: which agents a deployed orchestrator instantiates per task, with no competitive-analysis treatment that does not first fix a model of that response.

6.2 Conclusion

On a controlled proxy, not a deployed multi-agent system, a $2.6\times$ reduction produced an uncertain 4.2-point loss (95% CI -10 to $+1$), and an oracle reached $22\times$ with an observed gain — a ceiling plausibly an order of magnitude, not shown free in between. The obstruction is not degradation or item interaction (both occur) but that the seed-set object is off by a factor n from the per-receiver one on well-behaved instances, unboundedly once saturation is added, at no cost in approximability. That remaining gap points to relevance estimation, not allocation algorithms, as the bottleneck: switch to per-receiver allocation, worth 19 points over the seed-set alternative, then spend further effort on retrieval.

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